KPS PUBLIC SCHOOLS • STUDENT TECH BENCH PROGRAM

Lenovo 500e Chromebook

Solder-Free Repair Guide



Every field-replaceable repair that needs a screwdriver, not a soldering iron.
Lenovo 500e Chromebook, first generation (type 81ES): convertible with stylus.

In this guide

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Built from Lenovo's Hardware Maintenance Manual (500e Chromebook, January 2018).
Part numbers: look up by serial on pcsupport.lenovo.com before ordering.

1 Read first

This guide covers every Lenovo 500e Chromebook repair the Tech Bench can do with hand tools: swapping a part, reseating a cable, or fixing software. Anything that needs a soldering iron, hot air, or board-level work is in Section 6 and goes to the lead tech or the manufacturer.

How this guide was built. Screw counts, sizes and the order of removal come from Lenovo's Hardware Maintenance Manual for the 500e Chromebook (type 81ES). This guide rewrites Lenovo's procedures step by step for a student bench and adds tests and checks. Where Lenovo's manual doesn't give a count, the step says so and the first-teardown notes box is there to record it.

Don't use the Developer Mode battery steps. Lenovo's manual offers ways to switch the battery off in software. On district devices, unplug the battery connector instead: never enter Developer Mode.

Bench levels

Level	Who may do it	Typical repairs
LEVEL 1	Any trained student	Software fixes, single keys, bottom cover
LEVEL 2	Student with supervisor sign-off on this repair	Battery, Wi-Fi, touchpad, speakers, I/O board, display parts, hinges
LEVEL 3	Lead tech or adult supervisor only	Heat sink, upper case (keyboard) transplant, liquid damage, system board

Before you touch any device

1. ☐ Check the asset tag and look the device up in the district inventory. Note any open ticket.
2. ☐ Ask the lead tech whether the unit is still under Lenovo warranty or an accidental-damage plan. A warranty claim may cost less than a part, and opening a unit can affect warranty coverage.
3. ☐ Photograph all four sides of the device before opening it.
4. ☐ Record the symptom in the user's own words on the ticket.
5. ☐ Warn the user that a reset erases local files (Downloads). Cloud files in Drive are safe.

Safety rules

- **Power off, unplug, remove any SD card and cables,** then disconnect the battery before any other work inside.
- **Lithium batteries:** never puncture, bend, or pry against the battery. A swollen, hot, hissing, or smoking battery is a Level 3 stop. Move it to the metal battery bin or sand and get an adult.
- **ESD:** wear a grounded wrist strap on an ESD mat. Touch boards only by the edges.
- **Ribbon cables:** unlock every ZIF latch before sliding a ribbon out. A locked ribbon tears.
- **Glass:** wear safety glasses when removing a cracked screen.
- **Before power-on:** check for loose screws inside. Shake gently and listen. A loose screw can short the board.

Tools and supplies

Item	Use	Notes
Phillips #0 and #1	All screws	Lenovo lists PH0/PH1
Plastic pry tool	Cover latches, bezel	Never metal
Plastic spudger, opening picks, plastic tweezers	Clips, latches, ribbons	Only plastic touches connectors
ESD mat + wrist strap	Static protection	
Magnetic screw mat	Keeping screws in order	Screws are often different lengths
90%+ isopropyl alcohol + soft brush	Cleaning contacts, spill residue	
Kapton tape	Re-taping cables and connectors	
Recovery USB or SD card, 8 GB+	ChromeOS recovery	Chromebook Recovery Utility
Known-good USB-C charger	Testing power and charging	
Safety glasses	Cracked-screen removal	
Fire-safe battery bin or sand bucket	Damaged batteries	Required at every bench

Basic skills (start here if you're new)

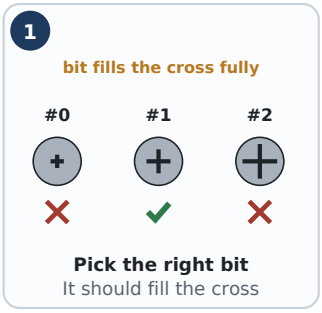
Every repair in this guide uses the same handful of skills. Learn these six first and the rest of the guide is just *which screws* and *which cable*. If a word in a repair is unfamiliar, look it up in the word list at the end of this chapter. Hard words in the steps are also explained in grey the first time they appear.

How a repair page works

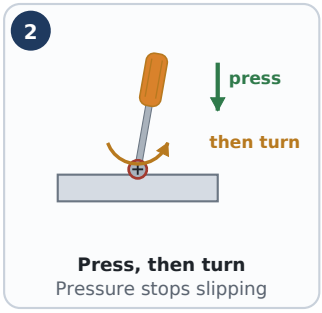
Part of the page	What it tells you
Coloured bar with the code (R5) and LEVEL	What the repair is and who may do it. Level 1: any trained student. Level 2: with a supervisor's sign-off. Level 3: lead tech only.
Fixes / Part / Time / Tools	Read this first. If you don't have the part or the tools, stop and get them before you open anything.
In plain words	One or two sentences on what you're about to do and why.
At a glance (numbered pictures)	The idea of the repair in four pictures. They show the idea, not your exact model.
Where it is on this model	A map of the inside with the part highlighted.
Step by step (☐ boxes)	The exact steps. Tick each box as you go so you never lose your place.
Common mistakes / If something goes wrong	The traps other students fall into, and what to do when a step doesn't work.
Verify before closing the ticket	The checks that prove the repair worked. Never skip them.
First-teardown notes	Write in what you found the first time (real screw counts, tape positions). It helps the next student.

Skill 1: Screws

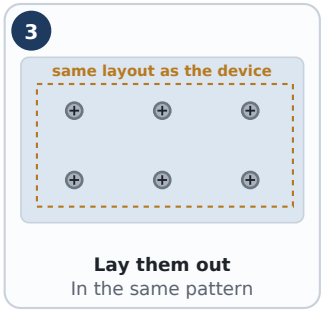
Most damage on a repair bench comes from screws: stripped heads, screws in the wrong hole, or a long screw driven through a board. Go slowly; screws are tiny and soft.



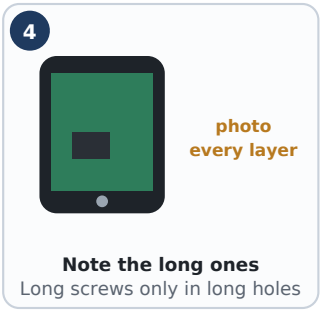
Pick the right bit
It should fill the cross



Press, then turn
Pressure stops slipping



Lay them out
In the same pattern

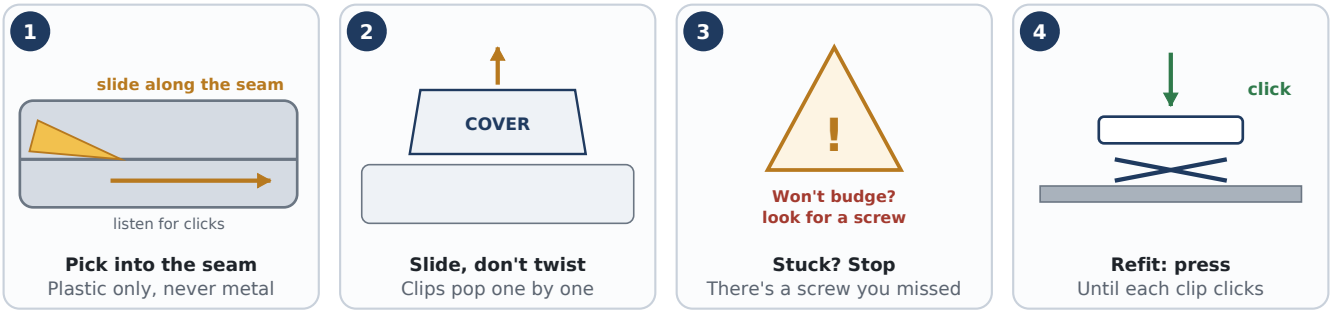


Note the long ones
Long screws only in long holes

- Most Chromebook screws take a **Phillips #0 or #1** bit. If the bit wobbles or slips, try the other size.
- Turn **left to loosen** (counter-clockwise), right to tighten. Tighten until it stops, then stop. "Snug" is enough: over-tightening cracks plastic.
- **Captive screws** stay in the cover. Loosen them until they spin freely; they won't come out.
- If a screw head starts to strip (the cross rounds off), stop and get the lead tech. Don't keep turning.

Skill 2: Opening clips

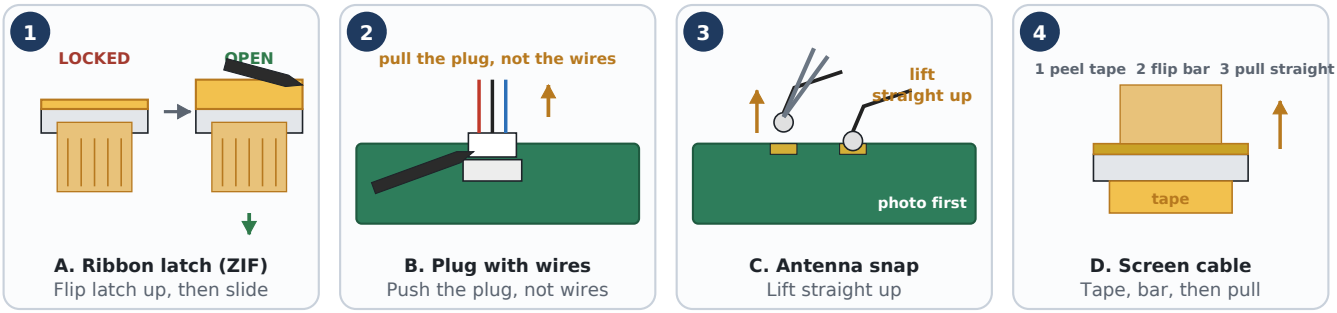
Covers and bezels are held by small plastic clips as well as screws. You release them by sliding a plastic pick along the seam, not by pulling.



- A click or pop is normal. A crack or creak means stop and look for a screw.
- Start where the guide says (usually a corner) and work around slowly.
- Never pry near the battery or against the screen glass.

Skill 3: The four kinds of connectors

Inside every Chromebook, parts plug into the main board in only four ways. Learn these and you can unplug anything safely.



Type	Looks like	How to unplug	How to plug back in
A. Ribbon latch (ZIF)	Flat, thin, see-through cable with gold stripes, in a slot with a small hinged flap	Flip the flap up with a fingernail or spudger. The ribbon slides out with no force.	Gold stripes facing the same way as before. Push it in straight and all the way, then close the flap.
B. Plug with wires	Small white plug with coloured wires (battery, speakers, fan)	Push the plug body out with a spudger, gently rocking. Never pull the wires.	Line it up, press straight in until it seats flat.
C. Antenna snap	Tiny round gold button on a thin black or white wire	Lift straight up with plastic tweezers or a fingernail.	Hold it right over the post and press straight down until it snaps.
D. Screen cable	Wider connector, often under tape, with a metal bar or pull tab	Peel the tape, flip up the metal bar (or pull the tab), then pull straight out.	Push straight in, flip the bar down, put the tape back.

Battery first, battery last. Unplug the battery (type B) before you unplug anything else, and plug it back in only after everything else is connected.

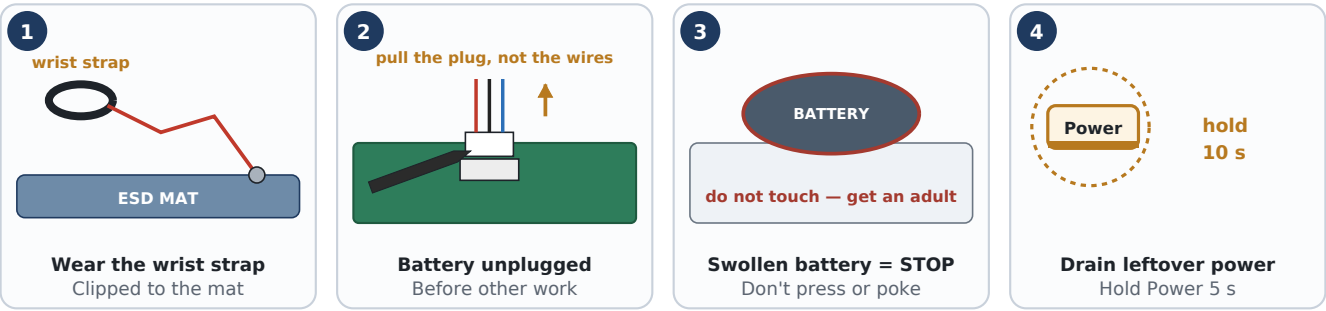
Skill 4: Tape, Mylar and adhesive

Many parts are held by tape as well as screws. Some tape is meant to be reused; some must be replaced.



- **Mylar** is thin plastic sheet that insulates parts. Put it back where it was: it stops short circuits.
- **Kapton** is thin amber heat-proof tape for holding cables down. It peels off and can be reused if clean.
- **Stretch-release tape** (under screens and batteries) lets go when you pull its tab slowly and flat. Pulling up snaps it.
- Conductive (shiny) tape on touchpads and cables must touch metal again when you refit, or the part acts strangely.

Skill 5: Static and battery safety



- Static from your body can kill a board without a spark you can see. Strap on, and touch boards only by the edges.
- A swollen, hot, hissing or sweet-smelling battery is an emergency: don't press it, move it to the fire-safe bin with an adult.
- Water, drinks and food stay off the bench.

Skill 6: Photograph, then test before you close



- Your photos are your instructions for putting it back together. Take more than you think you need.
- Plug in, power on and test the repaired part while the cover is still off. Finding a loose cable now saves taking it all apart again.
- Count your screws at the end. A leftover screw means something isn't fastened.

Word list

Word	What it means
Bezel	The plastic frame around the screen.
Bottom cover (base, D cover)	The panel underneath the Chromebook.
Top cover (palm rest, C cover)	The part your wrists rest on; often has the keyboard built in.
Lid / A cover	The back of the screen half.
Board / system board / mainboard	The main circuit board with the processor, memory and storage soldered on.
Daughterboard / I/O board	A small extra board holding some ports, joined to the main board by a ribbon cable.
Ribbon cable (FFC)	A flat, thin cable with gold stripes at the ends.
ZIF connector	A slot with a small flip-up latch that clamps a ribbon cable. "Zero insertion force": open it and the ribbon slides in or out freely.
eDP / LVDS cable	The video cable that carries the picture from the board to the screen.
Coax / antenna lead	The thin round wires from the Wi-Fi card to the antennas in the lid.
Spudger	A plastic or nylon pry stick. Safe to use near electronics.
Pick	A thin plastic triangle used to open seams and clips.
Captive screw	A screw that stays in the part after you loosen it.
Broad-head screw	A screw with a wide, flat head. Often used where it clamps plastic.
Mylar	Thin plastic insulating sheet.
Kapton tape	Thin amber heat-proof tape.
Thermal paste / pad	Grey paste or a soft pad that carries heat from the processor to the heat sink.
Heat sink	Copper or aluminium part that pulls heat off the processor.

Word	What it means
ESD	Electrostatic discharge: static electricity that can damage boards.
Isolator	A rubber piece that stops a part (usually a speaker) from rattling.
Powerwash	A factory reset of ChromeOS. Erases files saved on the device.
Recovery	Reinstalling ChromeOS from a USB stick or SD card.
EC reset (hard reset)	Restarting the small chip that controls power, charging and the keyboard.
Enroll	Connecting the Chromebook to the district's management so school policies apply.
eMMC	The storage chip on the board (like a small SSD). Soldered, so it can't be swapped.
Level 1 / 2 / 3	Who may do a repair: any trained student / student with sign-off / lead tech only.

2 Know the machine

Specs that matter for repair

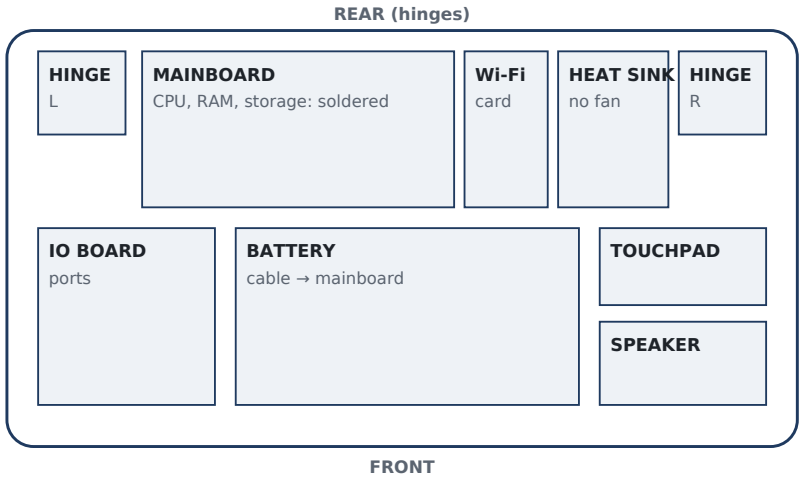
Item	500e	Why it matters
Processor	Intel Celeron N3450	Soldered
Memory / storage	4 GB RAM; 32 GB eMMC	Soldered
Display	11.6" HD IPS touch, 360° hinges	Stylus garage
Access	Bottom cover: 10 × M2.5×6	
Cameras	Front camera + world-facing camera	World-facing camera on the system board side
Wireless	Socketed Wi-Fi card, 1 screw	Replaceable

How it comes apart

A screwed-on bottom cover comes off first. Under it: battery, system board, heat sink, I/O board, speakers and touchpad. The keyboard is built into the upper case (palm rest), the last part left after everything else is removed.

Inside the base

With the cover off you see the parts below. It's a rough map; correct it on your first teardown.



ILLUSTRATIVE — block positions are approximate. Confirm on your first teardown.

500e parts list

Lenovo part numbers vary by configuration. Look up the exact part by serial number before ordering.

Part	Part number	Repair

Look up by serial. Lenovo's FRU numbers depend on the build. Use PartsLookup with the serial on the bottom label.

3 Triage: symptom → repair

Start with the cheapest, least invasive fix. Work through "try first" before ordering parts.

Symptom	Try first	Then	Part likely needed
Won't turn on, no lights	Known-good charger 30 min → S1	R4 → R3	Battery, or system board
Not charging	Other USB-C port and charger → S1	R3 → Sec. 6 if a port is dead	Battery or system board
Battery drains fast / low health in S2	S2 battery test	R3	Battery
Swollen battery / bulging case	STOP — Level 3	R3 by lead tech	Battery
Black screen, power light on	External display via USB-C; S1	R4 → R13 → R12	Display cable or panel
Flicker or lines that change with lid angle	S1	R13	Display cable
Cracked or blotchy screen	—	R12	LCD panel
Bezel loose or broken	Reseat clips	R11	Bezel
Lid floppy, hinge cracked	Check hinge screws	R15	Hinge(s)
One key missing or broken	—	R1	Keycap
Several keys or a row dead	S2 keyboard test; S1; S4	R4 → R16	Keyboard / top cover
Keys type wrong symbols	S4	S4	None
Touchpad dead or erratic	S1; check touchpad settings	R4 → R6	Touchpad or its cable
USB-A / SD card / audio dead	Try another device in the port; S1	R8	I/O board or its cable
Camera not detected	Privacy shutter open? S1	R14	Camera module
No sound (headphones work)	Volume, output; S1; S3	R7	Speakers
Wi-Fi missing or weak	S1 → S3	R5	Wi-Fi card or antennas
Runs hot, fan loud, shuts down	Check vents for lint; S2	R9	Fan or heat sink
Liquid spill	Power off NOW, unplug	R17	Depends on damage
"ChromeOS is missing or damaged"	S3 recovery	R18	System board (storage)
Cracked bottom or lid shell	—	R2	Cover

4 Software repairs

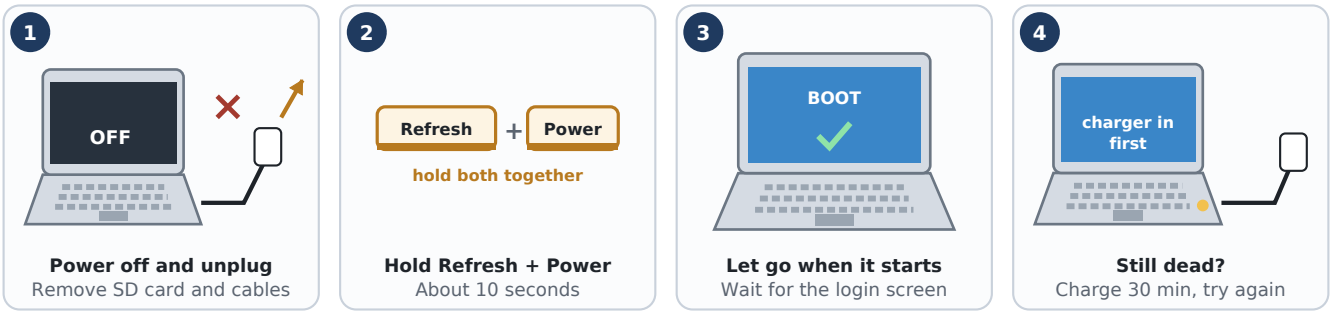
No tools needed. Always do S1 before opening a device.

S1 Hard reset (EC reset) LEVEL 1

Fixes	Won't power on, won't charge, frozen, black screen, unresponsive keyboard or touchpad
Time	2 minutes
Tools	Known-good USB-C charger
Data loss	None

In plain words. A hard reset restarts the small chip that runs power, charging and the keyboard. It's like unplugging something and plugging it back in, and it fixes a surprising number of "dead" Chromebooks. Nothing is erased.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

A hard reset restarts the embedded controller (power, keyboard and charging chip). It erases nothing.

Step by step

1. ☐ Unplug the charger and any USB devices. Remove any SD card.
2. ☐ Hold Power for 10 seconds so the device is fully off.
3. ☐ Hold **Refresh** (circular arrow), then tap **Power**. Release Refresh when it starts.
4. ☐ If nothing happens: plug in a known-good charger, wait 30 minutes, and repeat with the charger connected. Try the other USB-C port too.

Common mistakes
✗ Giving up after one try without the charger.
✗ Holding only Power (that's just a shutdown).
✗ Using a charger you haven't tested.

If something goes wrong
Nothing at all, even charging → Try the other USB-C port, then check the battery (battery repair).
Starts then shuts off → Run diagnostics (S2) for the battery.

Verify before closing the ticket

- Device reaches the login screen.
- Charging light comes on with the charger connected.

Tip. After a battery has been disconnected, plug in a charger before deciding the device is dead. Many Chromebooks won't wake on the power button alone.

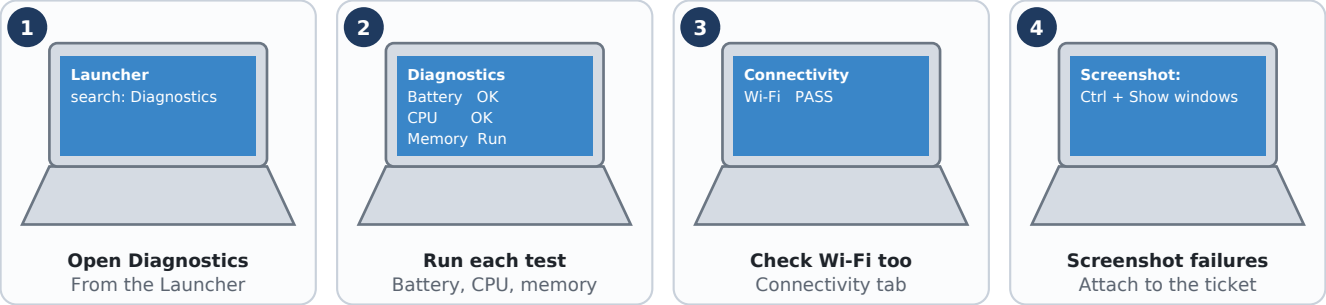
S2 Built-in hardware diagnostics

LEVEL 1

Fixes	Confirms the fault before ordering a part: battery, CPU, memory, keyboard, touchpad, Wi-Fi
Time	5–15 minutes
Tools	None
Data loss	None

In plain words. Chromebooks have a built-in checkup app. It tests the battery, processor, memory and more, and tells you what's failing before you order anything.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Step by step

1. ☐ Sign in with a bench test account if you can.
2. ☐ Open the Launcher, search **Diagnostics**, and open it.
3. ☐ **System:** read battery health and cycle count, then run the battery tests.
4. ☐ Run the CPU and memory tests.
5. ☐ **Connectivity:** confirm Wi-Fi is detected and run the tests.
6. ☐ **Input:** open the keyboard tester and press every key; test the touchpad and touchscreen.
7. ☐ Screenshot any failed result (Ctrl + Show windows) and attach it to the ticket.

Common mistakes

✗ Skipping the memory test because it's slow.
✗ Not saving the results to the ticket.
✗ Replacing a battery that tests healthy.

If something goes wrong

Battery health low → Battery repair.
Memory or storage fails → System board: Section 6 or lead tech.

Verify before closing the ticket

- Results recorded on the ticket.
- Parts ordered only for a failed test or a fault you saw yourself.

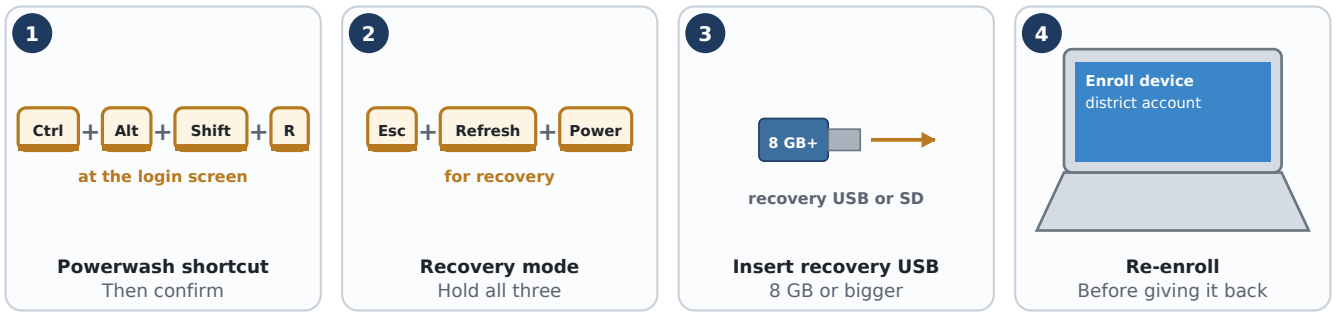
S3 Powerwash and USB recovery

LEVEL 1

Fixes	Slow, crashing, corrupted OS, "ChromeOS is missing or damaged", Wi-Fi missing after an update
Time	Powerwash 10 min; recovery 20–40 min
Tools	Recovery USB or SD card (8 GB+)
Data loss	**Yes.** Erases all local files (Downloads).

In plain words. A Powerwash wipes the Chromebook back to factory settings, like a fresh start. Recovery goes further and reinstalls ChromeOS from a USB stick. Both erase files saved on the device, so warn the student first.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

District-managed devices. Enrolled devices should force re-enrollment on their first Wi-Fi connection after a wipe. Confirm with the lead tech that forced re-enrollment is on and the bench Wi-Fi reaches enrollment. Never enter Developer Mode on a district device.

Step by step

1. ☐ **Check for updates first:** Settings > About ChromeOS > Check for updates.
2. ☐ **Powerwash (factory reset):** sign out, press Ctrl + Alt + Shift + R, choose Restart, then Powerwash. (May be blocked by policy; use recovery instead.)
3. ☐ **Make recovery media:** on a working computer, open the Chromebook Recovery Utility, choose the **Lenovo 500e Chromebook** from its list (or search the model), insert an 8 GB+ drive, and click Create now.
4. ☐ Hold **Esc + Refresh**, tap **Power**, and release when the recovery screen appears.
5. ☐ Insert the recovery media and follow the prompts.
6. ☐ After it reboots, connect to Wi-Fi and confirm it re-enrolls to the KPS domain.

Common mistakes
✗ Not warning that Downloads will be erased.
✗ Choosing Developer Mode (never on district devices).
✗ Handing it back before re-enrolling.

If something goes wrong
"ChromeOS is missing or damaged" again after recovery → Storage failing: system board, Section 6.
Can't enroll → Lead tech checks the Admin console.

Verify before closing the ticket

- Boots to the enrollment or KPS sign-in screen.

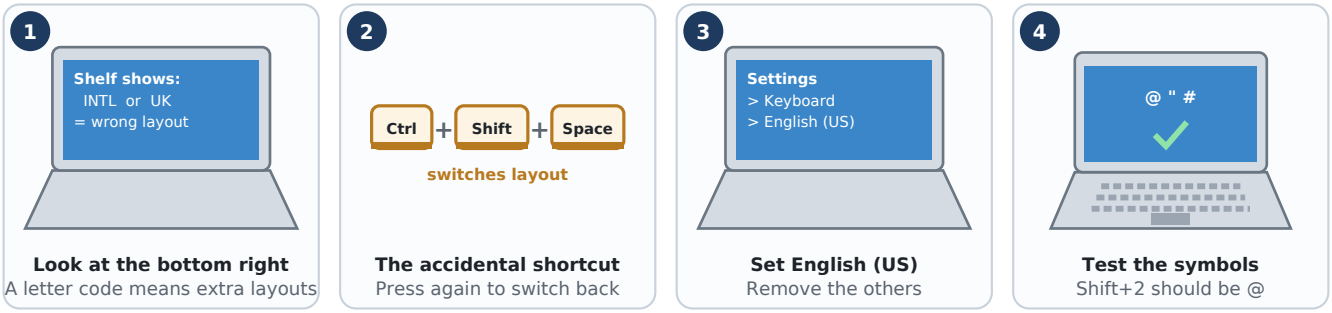
- Shows as enrolled in the Admin console.
- Recovery can't write to storage: storage has failed. That's a system board repair or warranty claim.

S4 Keyboard layout and international keyboard fix — LEVEL 1

Fixes	Keys type the wrong symbols (@ and " swapped, accents appear, Z and Y swapped); a replacement keyboard with a different layout
Time	5 minutes
Tools	None
Data loss	None

In plain words. If keys type the wrong symbols, the keyboard usually isn't broken: the computer has been switched to a different keyboard language. A key combination switches it by accident.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

ChromeOS can have several keyboard layouts installed. Students often switch layouts by accident with **Ctrl + Shift + Space**, or have **US International** on, which turns ' and " into accent keys. The physical keyboard is fine.

Step by step

1. ☐ Look at the bottom-right shelf. A layout code (US, INTL, UK, ES...) means more than one layout is installed.
2. ☐ Open **Settings > Device > Keyboard and inputs > Input** (on older versions: Settings > Advanced > Languages and inputs > Inputs).
3. ☐ Set **English (US)** as the active input. Remove any layouts the student doesn't need, especially US International, unless a teacher asked for it.
4. ☐ Test in a text box: Shift + 2 should type @ and Shift + ' should type ".
5. ☐ If a **replacement keyboard** has a different physical layout (UK keyboard on a US unit, for example), order the correct US part. Changing the software layout won't fix printed key labels.
6. ☐ If the wrong layout keeps coming back for everyone who signs in, it's set by district policy. Tell the lead tech; don't change it on the device.

Common mistakes

✗ Ordering a keyboard for a software problem.
✗ Leaving US International installed.
✗ Changing it when the district sets it by policy.

If something goes wrong

Wrong layout comes back → District policy sets it: tell the lead tech.
Printed keys are a different language → Wrong replacement part: order the US one.

Verify before closing the ticket

- Every key types its printed character in the S2 keyboard tester and in a text box.
- Only the needed layouts remain installed.

5 Hardware repairs

Most repairs start with the bottom cover (R2) and disconnecting the battery (R3).

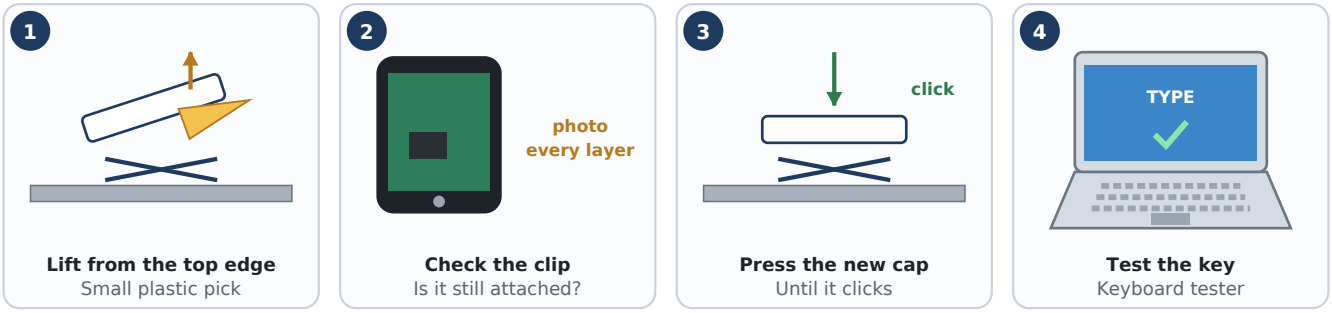
Golden rules. Photograph every layer before removing it. Lay screws on the mat in the same pattern as the device. Unlock every connector before pulling a cable. Never force a part: if it won't move, find the screw or clip you missed. Test before the last screws go in.

R1 Single keycap replacement — LEVEL 1

Fixes	Missing, cracked or loose key
Part	Individual key (ask the manufacturer or harvest from a scrapped keyboard)
Time	5 minutes
Tools	Plastic pick, tweezers

In plain words. A key is a plastic cap clipped onto a small scissor-shaped hinge. A missing or broken key can usually be fixed with just the cap, in a few minutes.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model

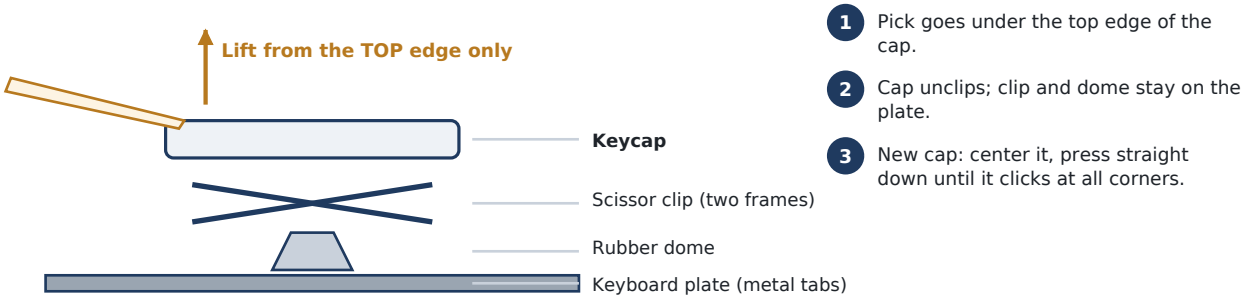


Illustration: side view of one key. Not to scale.

Step by step

1. ☐ Power off. Open the lid flat on the mat.
2. ☐ Slide a plastic pick under the TOP edge of the damaged key and lift until it unclips.

3. ☐ Check the scissor clip and rubber dome under it. A broken clip must be replaced too.
4. ☐ Fit a clip: hook its pins under the metal tabs on the keyboard plate, then press flat.
5. ☐ Center the new key over the clip and press straight down until it clicks at every corner.
6. ☐ Wide keys (Space, Shift, Enter): seat the metal stabilizer wire in its holders first.

Common mistakes
✗ Prying from the bottom edge (snaps the clip).
✗ Pressing a cap on crooked.
✗ Using a cap from a different model.

If something goes wrong
Clip is broken → Order the key with its clip, or replace the keyboard.
Key types nothing → Keyboard ribbon or keyboard: connectors repair first.

Verify before closing the ticket

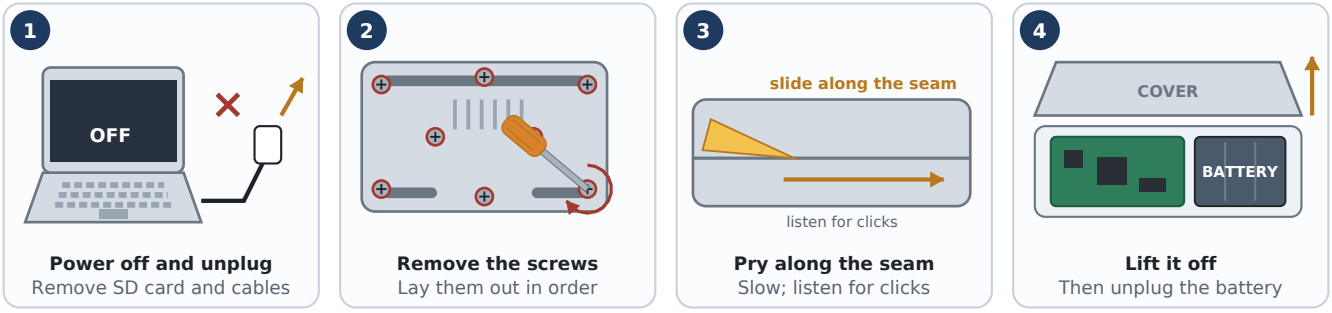
- Key clicks evenly and springs back.
- Key registers in the S2 keyboard tester.

R2 Base cover (bottom cover) removal — LEVEL 1

Fixes	Cracked or missing bottom cover; the first step for most internal repairs
Part	Bottom cover (look up by serial)
Time	10 minutes
Tools	Phillips driver, plastic picks, screw mat

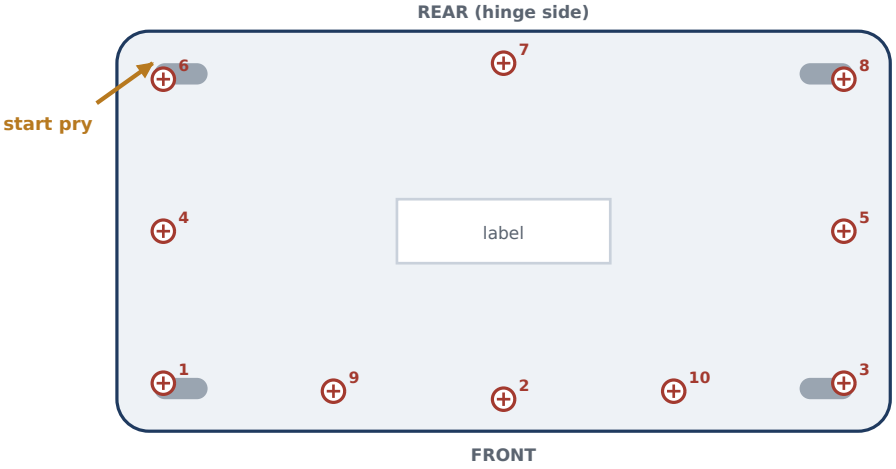
In plain words. The bottom cover is the plastic panel underneath. It's held by small screws and plastic clips. Taking it off is the first step for almost every repair inside.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Remove the 10 bottom screws.
2. Lay screws on the mat in this same pattern.
3. Pry from the corner shown, then work around the edge.
4. Disconnect the battery before anything else.

Illustration: underside. Positions illustrative; count from the service manual.

Step by step

1. ☐ Shut down fully, unplug, and remove any SD card and stylus.
2. ☐ Close the lid and turn the device upside down on the mat.
3. ☐ Remove the 10 M2.5×6 screws.
4. ☐ Release the cover with a plastic pick, starting at a rear corner, then work around the sides. Lift it off.
5. ☐ **Disconnect the battery now** (battery repair, first steps).
6. ☐ To refit: line up the cover, press around the edges until every latch snaps, then refit every screw. Snug only.

Common mistakes

✗ Missing a screw under a rubber foot and cracking the cover.
✗ Mixing up long and short screws.
✗ Using a metal tool in the seam (scratches, cracks).

If something goes wrong

Cover won't release in one spot → There's still a screw there. Check under the feet and labels.
A clip broke → It usually still holds. If it rattles, order a new cover.

Verify before closing the ticket

- No gaps around the seam; device sits flat.
- Every screw refitted, none left over.

First-teardown notes — fill in the first time you do this repair

Number of screws, and which holes take the longer ones?	_____

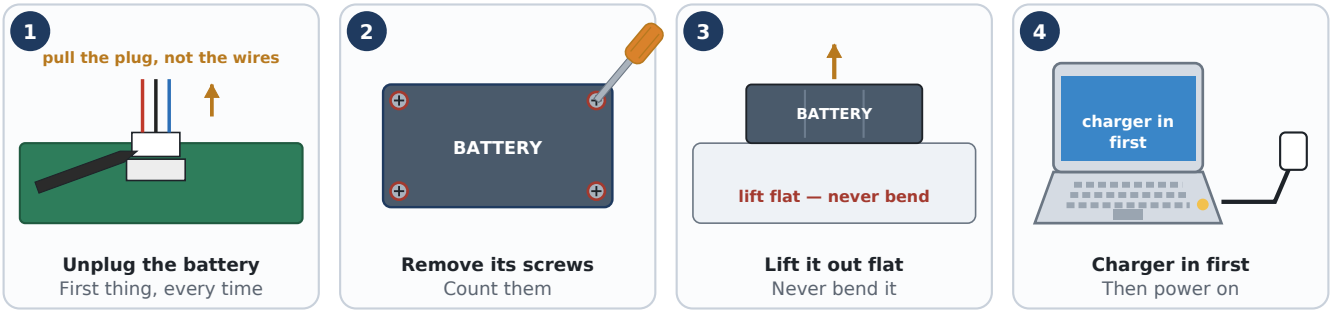
R3 Battery replacement

LEVEL 2

Fixes	Won't hold charge, low health in S2, swollen battery
Part	Battery (look up by serial)
Time	15–20 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, ESD strap

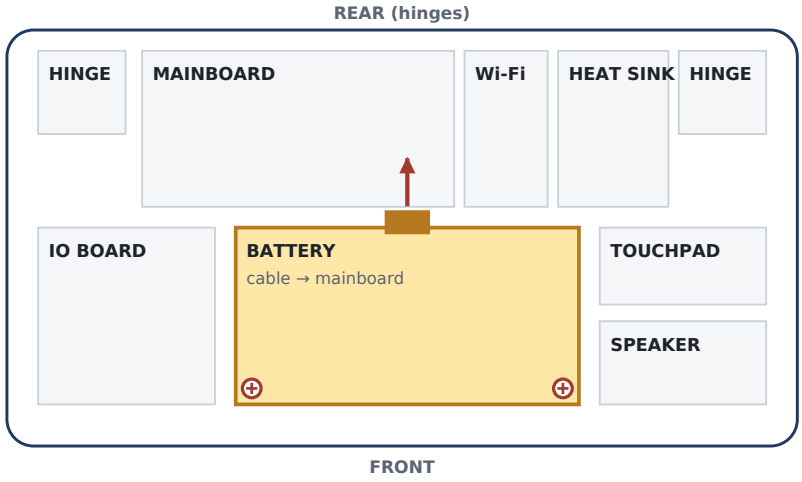
In plain words. The battery is a flat pack that powers the Chromebook when it's unplugged. To replace it you unplug its cable from the board, remove a few screws, and lift it out. Batteries can be dangerous if bent or punctured, so handle them gently.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Unplug the battery cable from the mainboard first (red arrow).
2. Pry at the plug body, never the wires.
3. Remove the battery screws (count them).
4. Lift out flat. Never bend or pry against the cell.

Illustration: base interior, battery highlighted. Screw positions illustrative.

Swollen battery = Level 3. Don't press or pry. The lead tech handles it; the old cell goes straight to the fire-safe bin.

No Developer Mode. Some service manuals disable the battery through Developer Mode or a key sequence. On district devices, unplug the battery connector instead.

Step by step

1. ☐ Remove the bottom cover.
2. ☐ Lift any tape over the battery connector and disconnect the battery cable from the system board.
3. ☐ Remove the 4 M2×4 silver screws.
4. ☐ Lift the battery out. Never bend it.
5. ☐ Fit the new battery, refit the screws, reconnect the cable.
6. ☐ Refit the cover. Plug in the charger before powering on.

Common mistakes
✗ Pulling the battery cable by its wires.
✗ Prying under the battery with a metal tool.
✗ Thinking it's dead because it won't start without the charger.

If something goes wrong
Swollen, hot, hissing or smells sweet → STOP. Don't press it. Fire-safe bin, get an adult.
Won't power on with new battery → Plug in the charger for 30 minutes, then hard reset (S1).

Verify before closing the ticket

- Charging light comes on; battery percentage shows.
- S2 battery health near 100% on a new battery.
- Old battery goes to battery recycling, never the trash.

R4 Reseating internal connectors — LEVEL 2

Fixes	Dead keyboard, touchpad or speakers; black screen; dead ports; no power after a drop
Part	None, unless a cable is damaged
Time	15–25 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, isopropyl alcohol

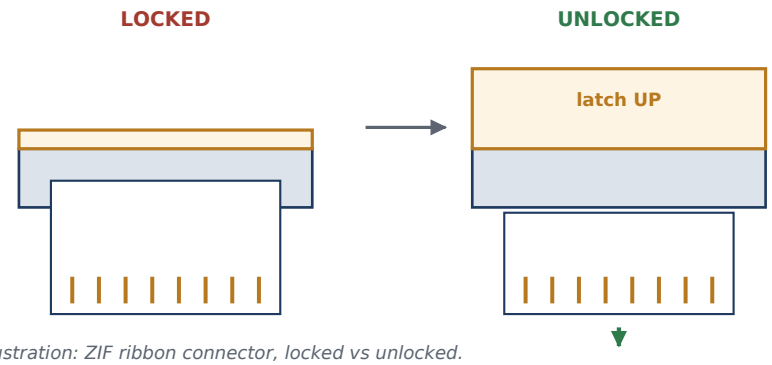
In plain words. Inside, parts connect to the main board with small plugs and flat ribbon cables. A drop can wiggle one loose. Unplugging it, checking it, and plugging it back in properly fixes many problems for free.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Flip the hinged latch up (or slide a drawer-type latch out) with a spudger.
2. Ribbon now slides out with zero force. Never pull a locked ribbon.
3. Refit: contacts facing the same way, push in straight and fully, close latch.
4. Board-to-wire plugs (battery, speaker): pry at the plug body, never the wires.

Illustration: ZIF ribbon connector, locked vs unlocked.

Connector	Type	How to release
Battery cable	Board-to-wire	Lift tape; pry at the connector body
Keyboard ribbon	ZIF ribbon	Unlock the latch; slide out
Touchpad ribbon	ZIF ribbon	Unlock the latch; slide out
Display (eDP) cable	Locking connector	Lift tape and the locking bar, pull straight
I/O board cable(s)	ZIF / FFC	Unlock both ends first
Speaker cable	Board-to-wire	Pry at the plug body
Wi-Fi antennas (2)	Snap-on coax	Lift straight up; white = Main, black = Aux

Step by step

1. ☐ Open the device and disconnect the battery.
2. ☐ Hold Power for 5 seconds with the battery disconnected to drain leftover power.
3. ☐ For each connector related to the symptom: photograph it, unlock and release it, and inspect it for bent pins, burn marks or creases.
4. ☐ Clean dirty contacts with a brush dampened in isopropyl alcohol. Let them dry 5 minutes.
5. ☐ Reconnect each one. Ribbons must go in straight and all the way before you close the latch.
6. ☐ Reconnect the battery last. Plug in and test **before** closing up.

Common mistakes
✗ Pulling a ribbon without unlocking the latch.
✗ Inserting a ribbon crooked or only half way.
✗ Forgetting to reconnect the battery last.

If something goes wrong
Ribbon is creased or torn → Replace that cable or part.
Symptom comes back later → The part itself is failing: replace it.

Verify before closing the ticket

- Symptom gone in a full test, not just at boot.
- Creased or torn ribbons replaced.

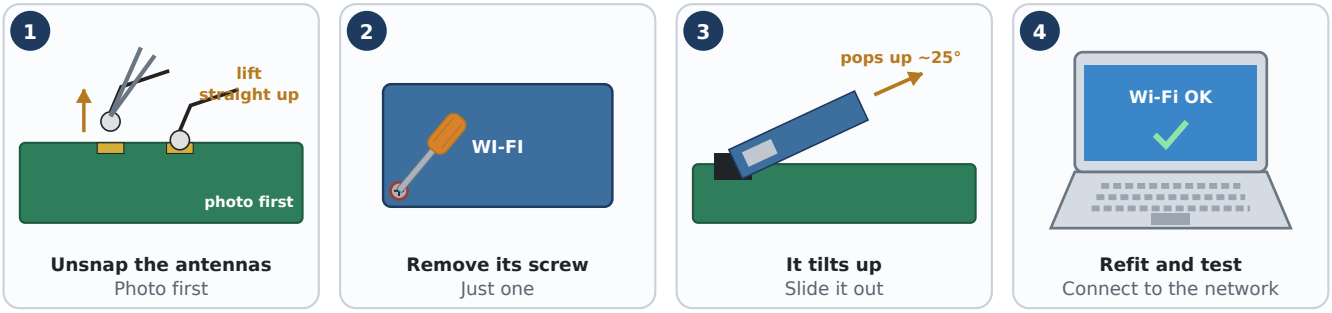
R5 Wi-Fi card reseat or replacement

LEVEL 2

Fixes	No Wi-Fi or Bluetooth, weak signal, Wi-Fi gone after a drop
Part	Wi-Fi card (look up by serial)
Time	15 minutes
Tools	Phillips driver, plastic tweezers

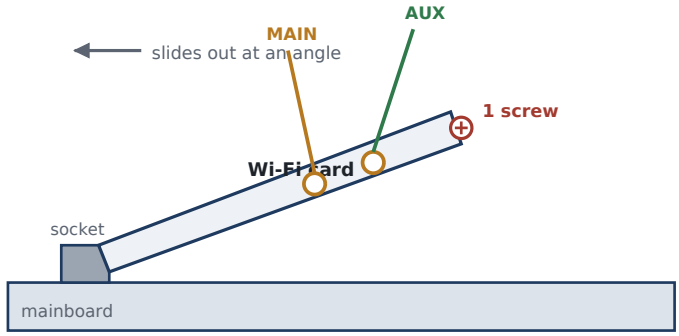
In plain words. The Wi-Fi card is a small removable card with two thin antenna wires snapped on top. It sits at an angle in a slot and is held down by one screw.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Photograph which antenna goes on which terminal.
2. Lift both antenna plugs straight up with plastic tweezers.
3. Remove the 1 screw. The card springs up at an angle.
4. Slide it out of the socket. Refit in reverse.

Illustration: M.2 Wi-Fi card, side view. Colors of antenna leads vary.

Step by step

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Find the Wi-Fi card.
3. ☐ Note which antenna goes where (white = Main, black = Aux), then lift both straight off.
4. ☐ Remove the 1 M2×3 screw.
5. ☐ The card tilts up. Slide it out of the socket at an angle.
6. ☐ Refit at an angle, press down, refit the screw, snap on both antennas and test.

Common mistakes
✗ Swapping the two antenna wires.
✗ Pulling antennas sideways (breaks the snap).
✗ Using a card that isn't the approved part.

If something goes wrong
Weak signal after a repair → An antenna wire is pinched or unplugged.
No Wi-Fi at all → Try a powerwash (S3) and a known-good card.

Verify before closing the ticket

- Wi-Fi connects and the S2 connectivity tests pass.
- Bluetooth finds a device.
- Weak signal after a lid repair usually means a pinched antenna lead.

Approved cards only. Use the Lenovo spare part for this model. Some firmware rejects other cards.

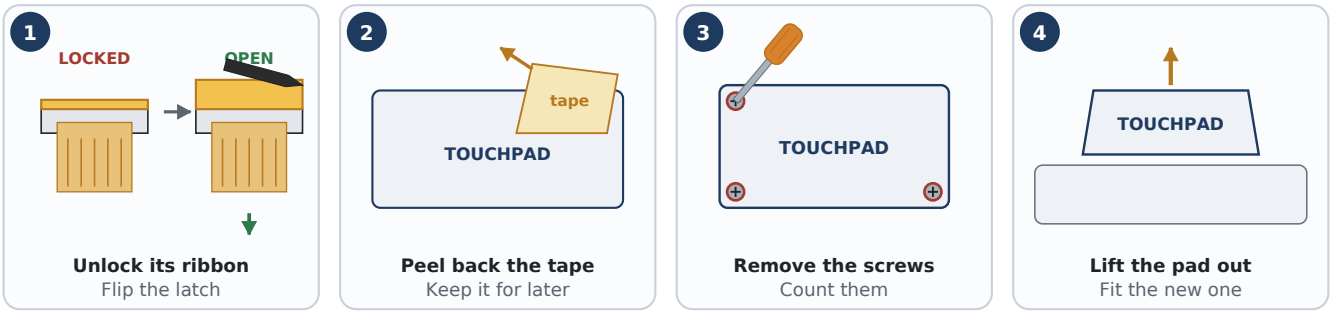
R6 Touchpad and touchpad cable replacement

LEVEL 2

Fixes	Touchpad dead, erratic, cracked or won't click (keyboard works)
Part	Touchpad (look up by serial)
Time	20–30 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, Kapton tape

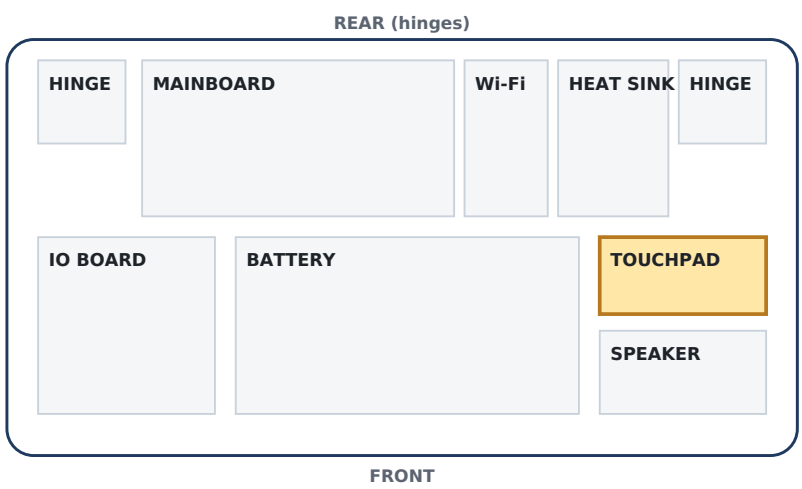
In plain words. The touchpad is a flat glass or plastic pad held from underneath by a few small screws and a bracket. A ribbon cable connects it to the board.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Battery disconnected first.
2. Unlock the ZIF latch before sliding the ribbon out.
3. Lift tape or Mylar; remove the screws.
4. Conductive tape must touch the pad on reassembly.

Illustration: base interior, touchpad highlighted. Position illustrative.

Check first: touchpad turned on in Settings, S1 done, and the cable reseated.

Step by step

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Unlock the ZIF (a flip-up latch that clamps a flat ribbon cable) latch and slide the touchpad ribbon out.
3. ☐ Lift any Mylar (a thin plastic sheet) or tape, then remove the touchpad bracket screws (count them; record the number in the notes box).
4. ☐ Lift the touchpad out.
5. ☐ Fit the new touchpad, refit the screws and tape, reconnect the ribbon and lock the latch.

Common mistakes

✗ Forgetting the conductive tape (the pad acts strange).
✗ Over-tightening screws (click stops working).
✗ Ribbon not locked.

If something goes wrong

Click doesn't work → Loosen the bracket screws a little.
Cursor jumps around → Check the tape and ribbon; try S3.

Verify before closing the ticket

- Cursor moves, clicks and two-finger scrolls.
- Click works at all four corners.
- Passes the S2 touchpad test.

First-teardown notes — fill in the first time you do this repair

Touchpad screw count and any tape positions. _____

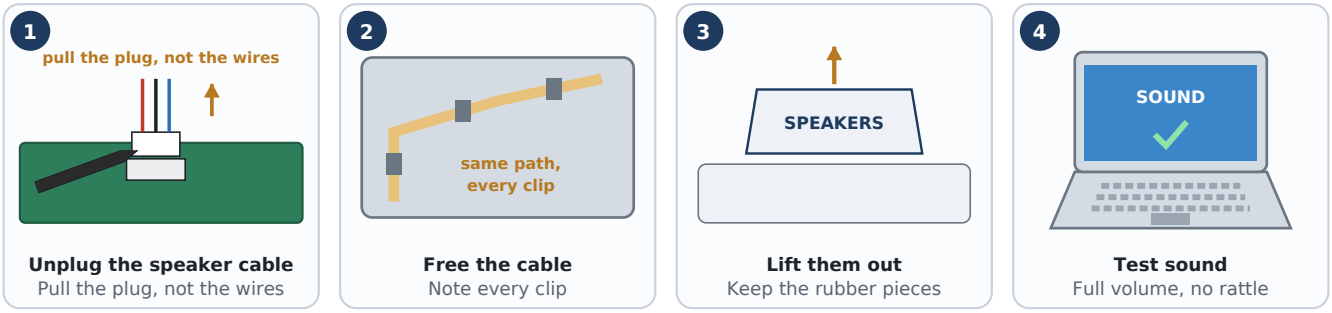
R7 Speaker replacement

LEVEL 2

Fixes	No sound or crackling from the speakers, with headphones working
Part	Speakers, left and right pair (look up by serial)
Time	20 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers

In plain words. The speakers are two small boxes joined by one cable. They unplug from the board, unroute from a few clips, and lift out. The rubber pieces stop them rattling, so keep them.

At a glance



Unplug the speaker cable
Pull the plug, not the wires

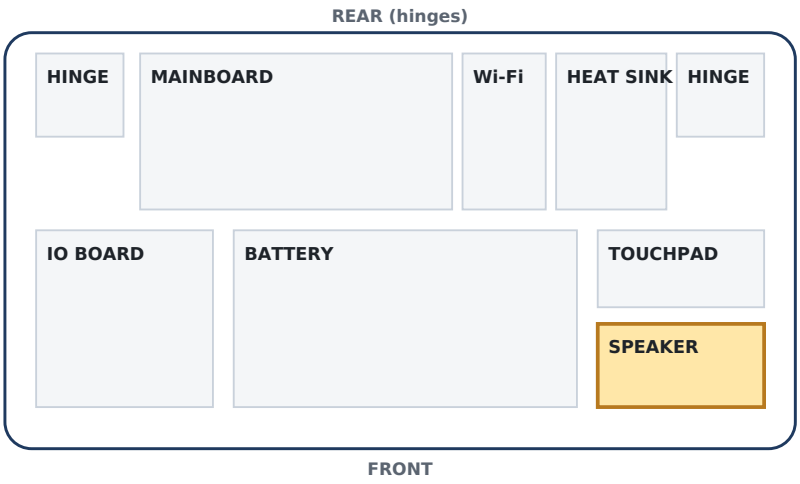
Free the cable
Note every clip

Lift them out
Keep the rubber pieces

Test sound
Full volume, no rattle

The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Battery disconnected.
2. Unplug the speaker cable at the board.
3. Unroute the cable, noting each clip.
4. Keep the rubber isolators: they stop rattle.

Illustration: base interior, speaker highlighted. Position illustrative.

Check first: if headphones also have no sound, the fault is software or the system board, not the speakers.

Step by step

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Disconnect the speaker cable from the system board and free it from its routing clips.
3. ☐ Remove the 4 M2×2.5 screws.

4. ☐ Lift both speakers out together, keeping the rubber isolators (rubber anti-rattle pieces).
5. ☐ Fit the new pair, route the cable exactly as before, reconnect.

Common mistakes
✗ Losing the rubber isolators (speakers buzz).
✗ Pinching the cable under the cover.
✗ Replacing speakers when headphones are silent too.

If something goes wrong
Headphones also silent → Software or board: S1, S3, then lead tech.
Rattle at high volume → Missing isolator or loose screw.

Verify before closing the ticket

- Clear sound from both sides with no rattle at full volume.
- Headphone jack still works.

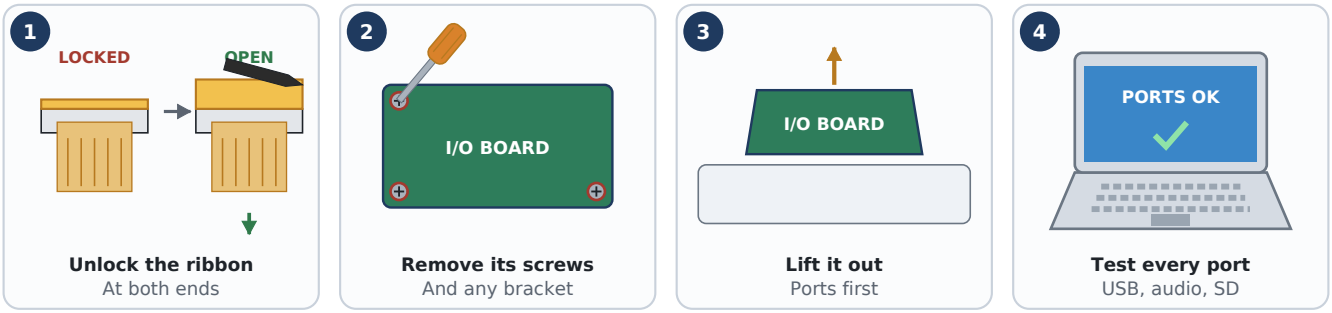
First-teardown notes — fill in the first time you do this repair	
Speaker screw count.	_____

R8 I/O (USB) daughterboard replacement LEVEL 2

Fixes	Dead USB-A, audio jack with the system board working
Part	I/O (USB) board and cable (look up by serial)
Time	20–30 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers

In plain words. The I/O board is a small separate board that holds some of the ports (like the USB-A port or headphone jack). It connects to the main board with a ribbon cable, so it can be swapped by itself.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model

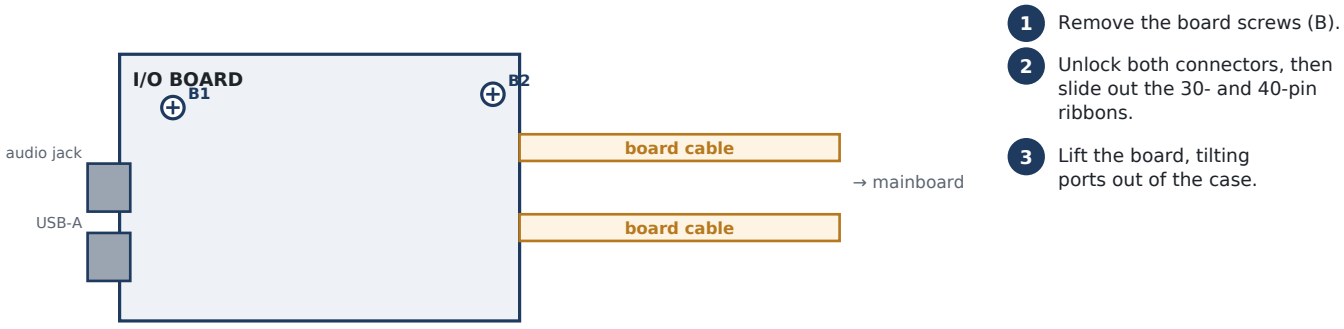


Illustration: USB board from above. Port order and screw spots illustrative. Confirm on yours.

Step by step

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Remove the display assembly first (Lenovo's order).
3. ☐ Remove the 4 M2×4 screws.
4. ☐ Disconnect the I/O cable and lift the board out.

Common mistakes

✗ Ribbon in upside down or crooked.
✗ Forgetting the bracket.
✗ Testing only one port.

If something goes wrong

Ports still dead → Ribbon cable or the main board connector.
USB-C dead → Usually on the main board: Section 6.

Verify before closing the ticket

- Every port on the board works (flash drive, charger, SD card, headphones as fitted).
- Bracket and all screws refitted.

First-teardown notes — fill in the first time you do this repair

I/O board screw count and what ports it carries. _____

R9 Heat sink replacement

LEVEL 3

Fixes	Overheating, thermal shutdowns, loud or rattling fan
Part	Heat sink (look up by serial)
Time	20–30 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, isopropyl alcohol, lint-free cloth

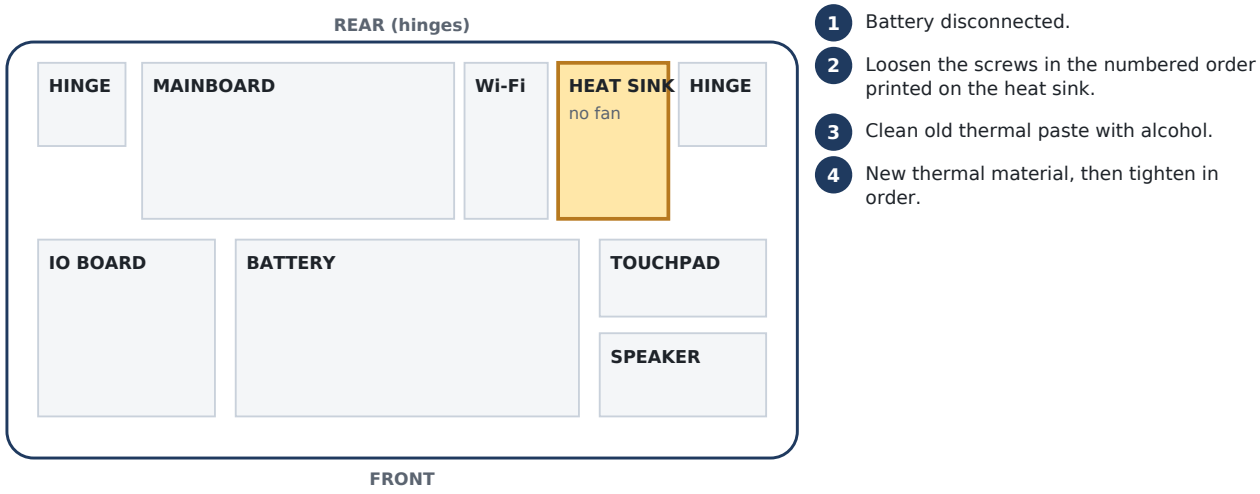
In plain words. The heat sink is a copper plate that pulls heat off the processor. A thin layer of thermal paste between them helps heat move. Whenever the heat sink comes off, the old paste must be cleaned off and new paste put on.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Battery disconnected.
2. Loosen the screws in the numbered order printed on the heat sink.
3. Clean old thermal paste with alcohol.
4. New thermal material, then tighten in order.

Illustration: base interior, cooling highlighted. Position illustrative.

Thermal material. Clean old thermal material off the processor and heat sink every time, and fit new material. Never run the board without the heat sink.

Step by step

1. ☐ Remove the bottom cover and disconnect the battery. On this model the heat sink comes off with the system board out; follow the board repair, then flip it over.

2. ☐ Loosen the heat sink screws (count them; record the number in the notes box) in the order printed on the heat sink.
3. ☐ Lift off the heat sink. Clean all old thermal material (heat-transfer paste) off the processor and heat sink.
4. ☐ Apply new thermal material, then refit, tightening in order.

Common mistakes
✗ Reusing old paste.
✗ Using way too much paste.
✗ Tightening one corner all the way first (cracks the chip).

If something goes wrong
Still overheating → Paste missing or uneven; fan blocked or dead.
Fan rattles → Replace the fan.

Verify before closing the ticket

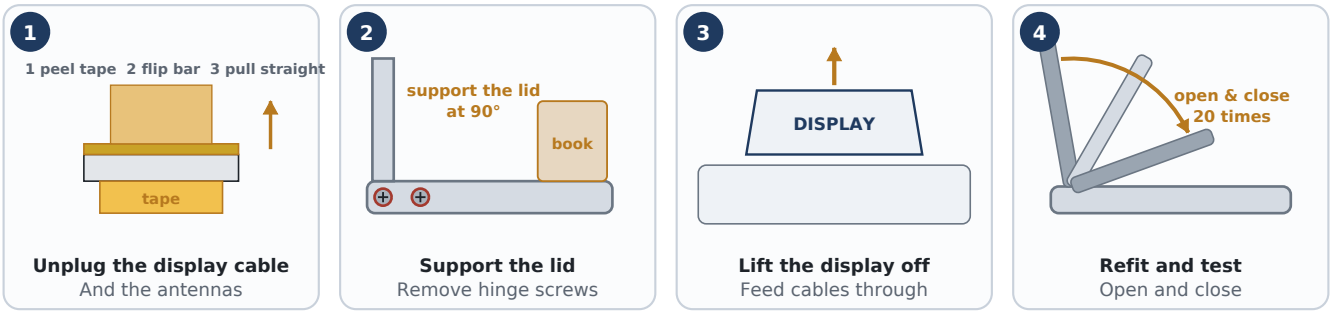
- Device runs a 10-minute video without shutting down.
- Fan (if fitted) spins up under load and is quiet at idle.

R10 Display assembly removal (separating lid from base) LEVEL 2

Fixes	Needed for hinge, lid and some board repairs; replacing a whole display assembly
Part	Display assembly or individual display parts (look up by serial)
Time	20–30 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers

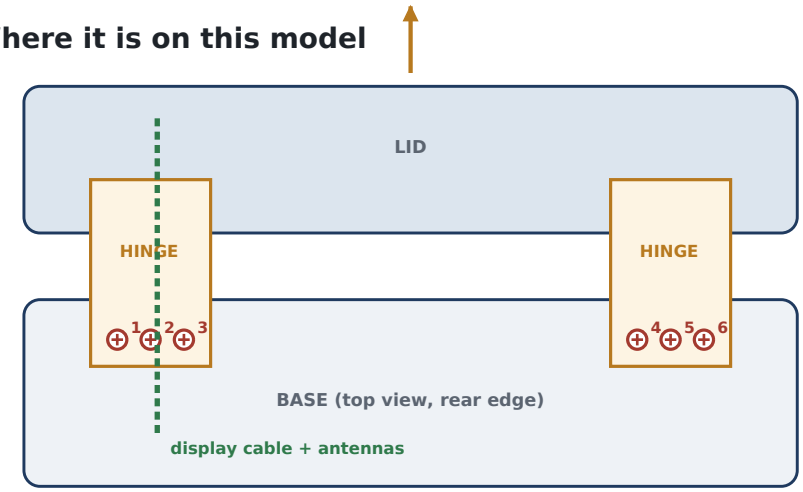
In plain words. The display (the whole lid with the screen) is attached to the base by hinges. To take it off you unplug the display cable and antenna wires, hold the lid up, and remove the hinge screws.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Battery out, display cable unplugged at the mainboard.
2. Unthread cable and Wi-Fi antennas from the guides FIRST.
3. Remove the display-to-base hinge screws (count them).
4. Support the lid, then lift it away from the base.

Illustration: hinge area seen from above with the D cover off. Positions approximate.

Step by step

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Disconnect the display cable (lift the tape and locking bar) and both Wi-Fi antennas; free them from their routing.
3. ☐ Open the lid to 90° or more and support it. Remove the 6 M2.5×5 hinge screws.
4. ☐ Lift the display assembly away from the base.
5. ☐ Refit in reverse; route the antennas and display cable back through every clip.

Common mistakes

✗ Letting the lid fall when the last screw comes out.
✗ Pinching a cable in the hinge on refit.
✗ Forgetting the antenna wires.

If something goes wrong

Weak Wi-Fi after refit → Antenna pinched or unplugged.
Screen flickers → Display cable not fully seated.

Verify before closing the ticket

- Display, camera and Wi-Fi all work after reassembly.
- Lid opens through its full range without pinching cables.

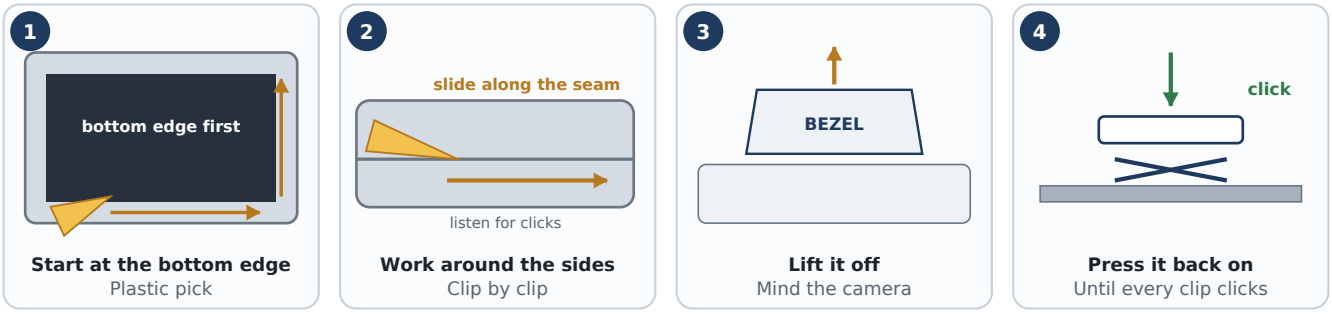
R11 Display bezel replacement

LEVEL 2

Fixes	Cracked, loose or missing screen bezel; the first step for panel, camera and display-cable repairs
Part	Display bezel (look up by serial)
Time	10 minutes
Tools	Plastic picks, Phillips driver

In plain words. The bezel is the plastic frame around the screen. It clips (and sometimes glues) onto the lid. You remove it to get to the screen, camera or display cable.

At a glance



Start at the bottom edge
Plastic pick

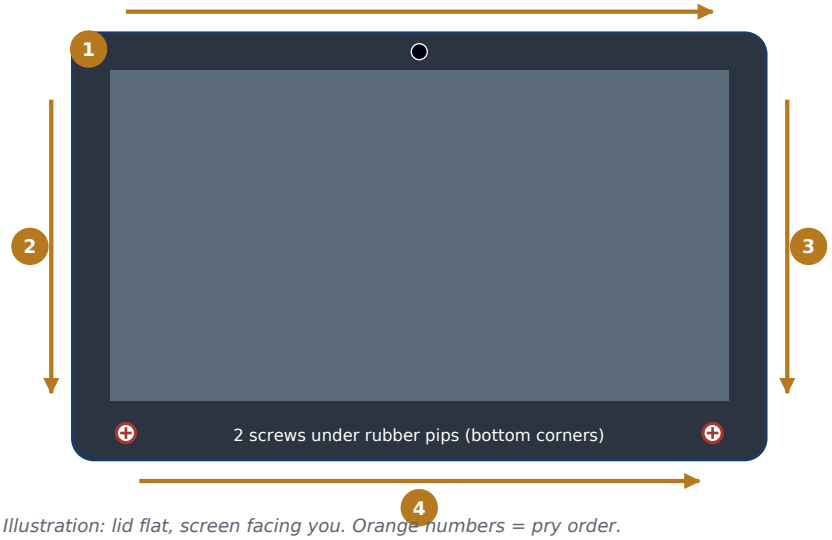
Work around the sides
Clip by clip

Lift it off
Mind the camera

Press it back on
Until every clip clicks

The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Remove the bezel screws first (under rubber caps if fitted).
2. Start prying at a corner with a flat pick.
3. Work down each side, then across the top.
4. Bottom (hinge) edge last.
5. Don't pull on the camera window.

Illustration: lid flat, screen facing you. Orange numbers = pry order.

Step by step

1. ☐ Open the lid flat.
2. ☐ Peel the rubber caps off the strip cover and remove the 2 M2×3 screws under them.
3. ☐ Pry the strip cover and bezel (the plastic frame around the screen) off with a plastic tool.
4. ☐ Refit and press until every clip seats.

Common mistakes
✗ Prying against the screen (cracks it).
✗ Bending the bezel too far.
✗ Forgetting new adhesive on glued bezels.

If something goes wrong
Gap after refitting → A clip isn't seated or broke.
Camera blocked → Bezel isn't lined up.

Verify before closing the ticket

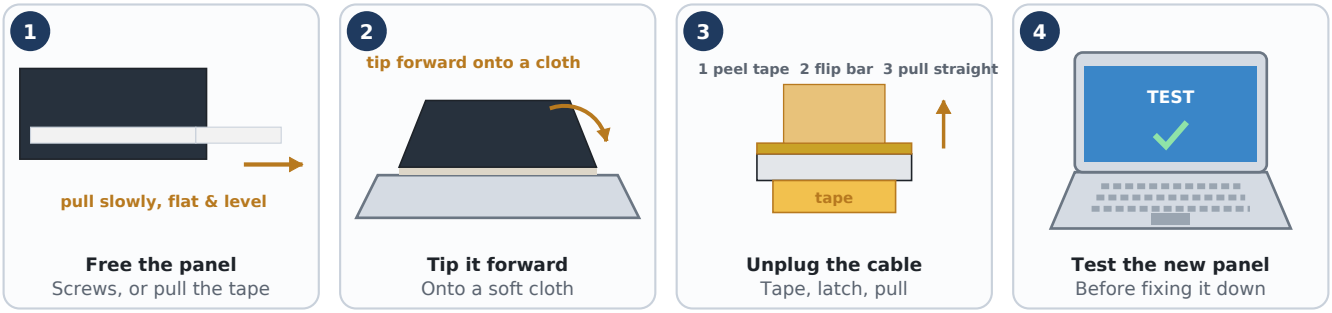
- Bezel sits flush all round.
- Camera window lined up; privacy shutter slides freely.
- Lid closes fully.

R12 LCD panel replacement — LEVEL 2

Fixes	Cracked screen, dead pixels, blotches, black screen not fixed by the display cable
Part	LCD panel (match the part number on the back of your panel)
Time	25–35 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, safety glasses, Kapton tape

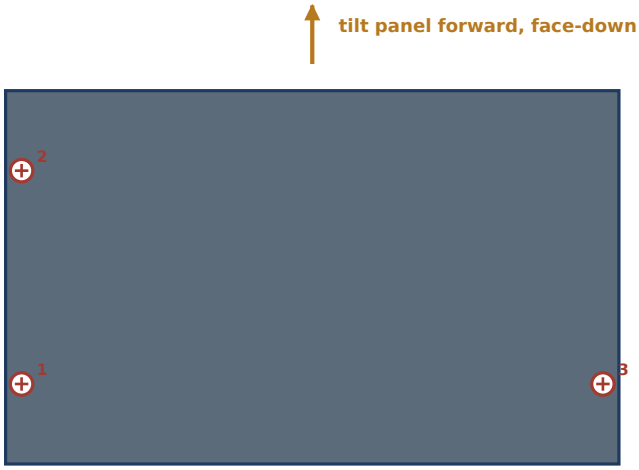
In plain words. The LCD panel is the screen itself. Once the bezel is off, it's held by small screws or stretchy tape. It tips forward so you can unplug its cable from the back.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Remove the 3 panel screws (sides of the panel).
2. Lay the panel face-down on the keyboard (cloth under it).
3. On the back: peel the tape over the connector, open the clip, pull the cable straight out.
4. Connect the new panel and TEST before screwing it down.

Illustration: lid with bezel removed. Screw positions are typical. Count them on yours.

Cracked glass. Wear safety glasses. Tape across a shattered panel before handling it.

Test first: if an external USB-C display works, the system board is fine and the fault is the panel or its cable.

Touch display. On a convertible the touch layer and glass are usually one unit with the panel. Order the touch panel part, not a plain LCD.

Step by step

1. ☐ Remove the bezel (the plastic frame around the screen).
2. ☐ Remove the 3 (or 4) M2×3 panel screws.
3. ☐ Tip the panel forward and disconnect the LCD cable and the sensor cable.
4. ☐ Connect the new panel and test the picture **before** fixing it down.
5. ☐ Refit everything in reverse.

Common mistakes

✗ Pressing on the screen face.
✗ Pulling the cable before opening the latch.
✗ Ordering the wrong panel (touch vs non-touch).

If something goes wrong

New panel also black → Display cable or board: cable repair next.
Touch not working → Wrong panel type or touch cable not connected.

Verify before closing the ticket

- Full image: no lines, dead pixels or uneven backlight.
- Brightness works across its whole range.
- Image steady through full lid travel.

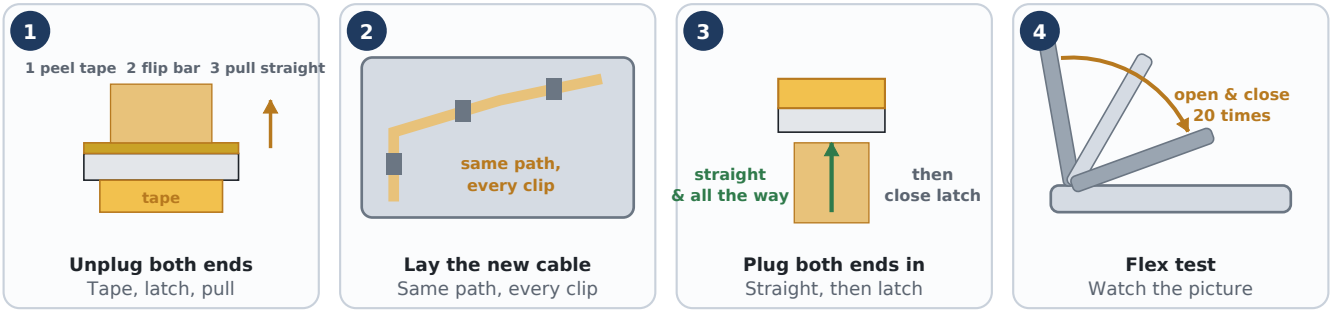
R13 Display (eDP) cable replacement

LEVEL 2

Fixes	Flicker, lines or black screen that changes with lid angle; external display works
Part	Display (eDP) cable (look up by serial)
Time	40–50 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, Kapton tape

In plain words. The display cable carries the picture from the main board up through the hinge to the screen. Hinges bend it every time the lid moves, so it wears out. Symptoms change when you move the lid.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model

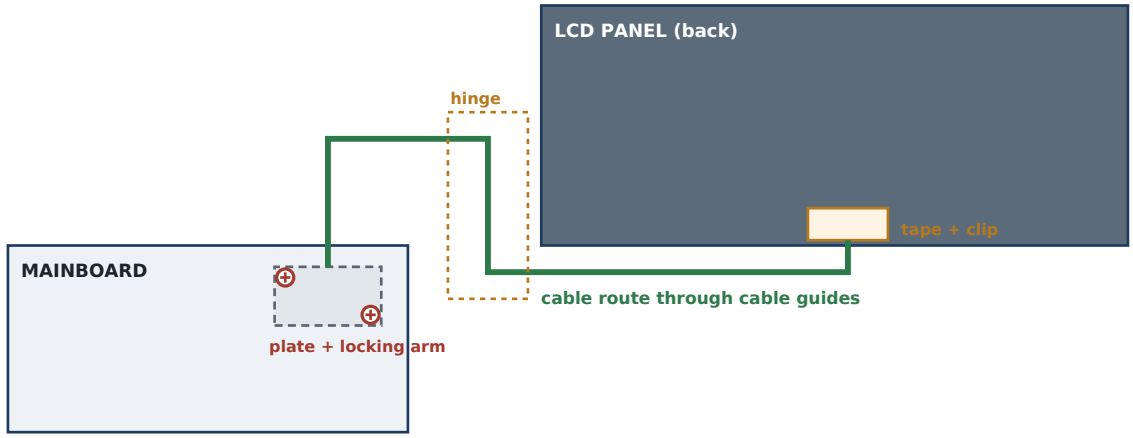


Illustration: display cable path, mainboard to panel (route simplified). Photograph the real route before removing.

Step by step

1. ☐ Remove the bezel (the plastic frame around the screen) and panel; disconnect the display cable at the system board.
2. ☐ Free the cable from the back cover and hinge area; note every clip and tape point.
3. ☐ Route the new cable exactly along the old path and reconnect both ends.
4. ☐ Test at every lid angle before reassembly.

Common mistakes
✗ Routing the cable a new way (it gets pinched).
✗ Leaving the tape off the connector.
✗ Not testing at different lid angles.

If something goes wrong
Flicker only at some angles → Cable still pinched or not seated.
Black screen, external display works → Panel: LCD panel repair.

Verify before closing the ticket

- Steady image at every angle: open and close the lid 20 times while watching.
- Camera still works if it shares the cable.

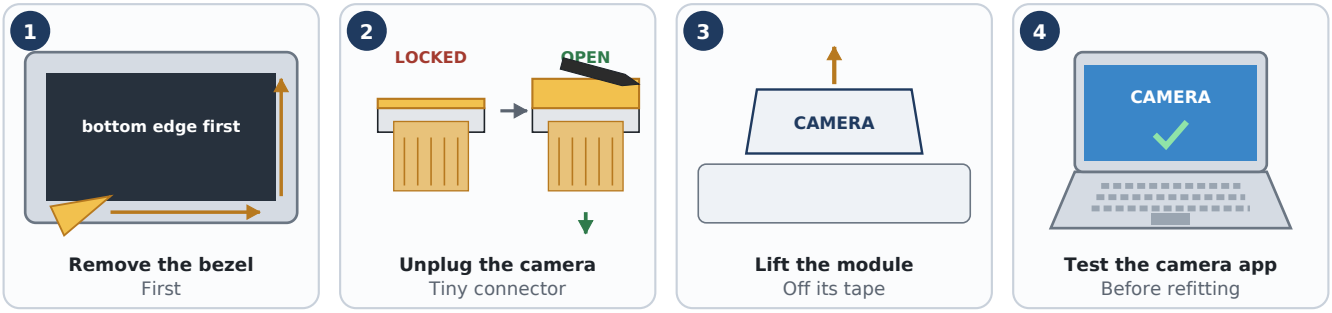
R14 Camera module replacement

LEVEL 2

Fixes	Camera not detected, black image, broken lens
Part	Camera module (look up by serial)
Time	15–20 minutes
Tools	Plastic picks, plastic tweezers

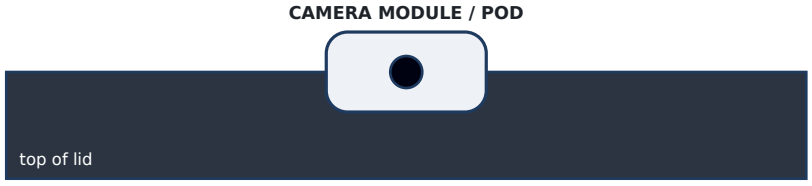
In plain words. The camera is a small strip at the top of the screen. It's usually held by sticky tape and connected by a tiny cable. Behind the bezel, it lifts out easily.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Bezel off first (it covers the camera's front).
2. Unplug the small camera cable with tweezers.
3. Lift the module by its board edges, never the lens.
4. Test in the Camera app before closing up.

Illustration: camera at the top-center of the lid. Cap shapes simplified.

Check first: privacy shutter open? Camera works after S1 and a powerwash (S3)?

Step by step

1. ☐ Remove the bezel (the plastic frame around the screen).
2. ☐ Lift any Mylar (a thin plastic sheet) or tape over the camera module, disconnect its cable, and lift it off (it's usually held by adhesive).
3. ☐ Fit the new module, reconnect, and test before refitting.

Common mistakes
✗ Touching the lens.
✗ Forgetting the privacy shutter is closed.
✗ Bending the camera strip.

If something goes wrong
Still not detected → Check the cable; try S3.
Image blurry → Lens dirty or fingerprinted.

Verify before closing the ticket

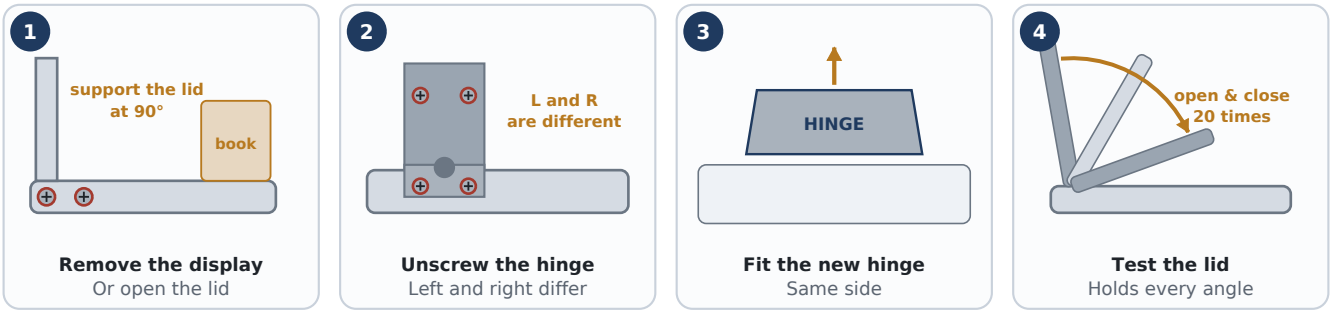
- Camera app shows a clear image.
- Privacy shutter fully covers the lens when closed.
- Works in a Google Meet test call.

R15 Hinge replacement — LEVEL 2

Fixes	Floppy lid, cracked hinge, grinding, lid won't stay up
Part	Hinges, left and right (look up by serial)
Time	30–45 minutes
Tools	Phillips driver, plastic picks

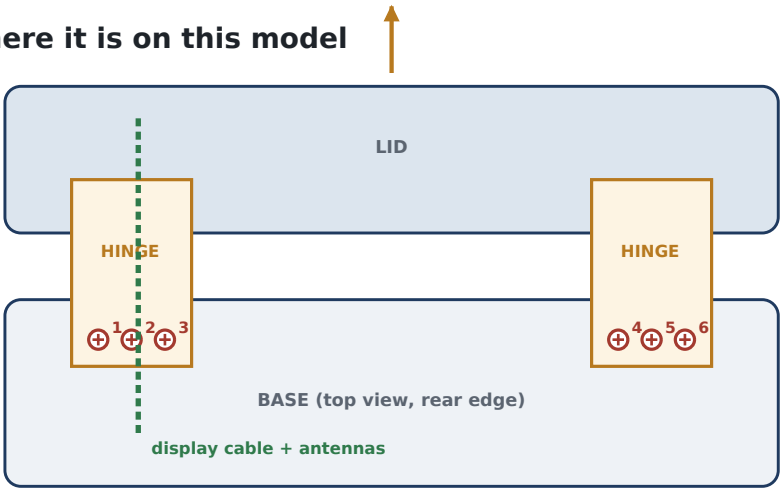
In plain words. Hinges are metal brackets that hold the lid to the base and keep it at any angle. A cracked or loose hinge makes the lid floppy.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Battery out, display cable unplugged at the mainboard.
2. Unthread cable and Wi-Fi antennas from the guides FIRST.
3. Base: 6 M2.5×5. Lid side: 6 M2×3 + 2 M2×2.5.
4. Support the lid, then lift it away from the base.

Illustration: hinge area seen from above with the D cover off. Positions approximate.

Check first: tighten the hinge screws. A screw that spins without tightening means a stripped post: that's a cover replacement.

Step by step

1. ☐ Separate the display assembly from the base (display assembly repair).
2. ☐ Remove the 6 M2×3 screws and 2 M2×2.5 screws.
3. ☐ Lift the hinges off the lid.
4. ☐ Fit the new hinges (left and right differ) and reassemble.

Common mistakes

✗ Mixing up left and right hinges.
✗ Pinching the display cable in the hinge.
✗ Forcing a stripped screw.

If something goes wrong

Screw spins and never tightens → Stripped post: cover replacement.
Lid floppy on one side → Wrong hinge or loose screw.

Verify before closing the ticket

- Lid holds any angle through its full travel.
- Both hinges feel equally stiff.
- Display steady and Wi-Fi strength normal (no pinched cables).

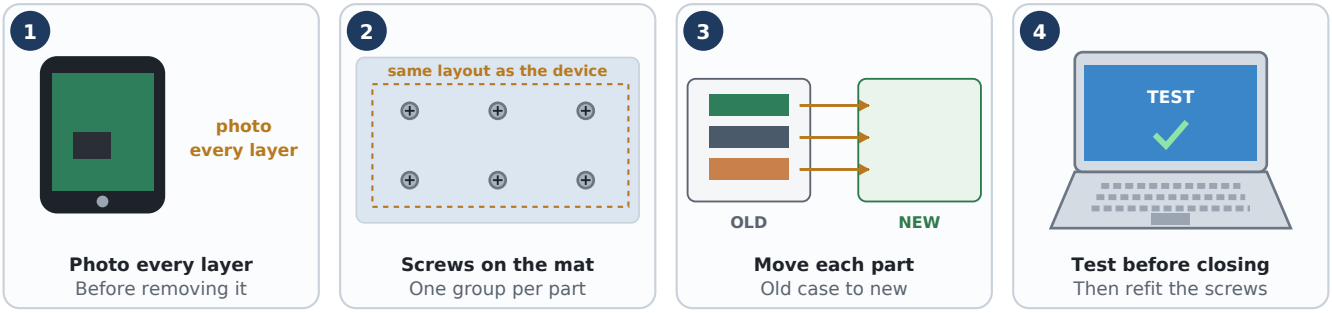
R16 Keyboard / top cover replacement

LEVEL 3

Fixes	Several dead keys, a dead row, liquid-damaged keyboard, cracked palm rest
Part	Upper case with keyboard (by keyboard language; look up by serial)
Time	60–90 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, ESD strap, screw mat, Kapton tape

In plain words. On this model the keyboard is built into the top case (the palm rest). To replace it, every other part moves from the old case to the new one. It's a big job: take photos and go slowly.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



Illustration: what moves to the new top assembly. Check what comes pre-fitted first.

The keyboard is part of the upper case (palm rest). Everything else in the base moves to the new upper case: this is a full transplant.

Step by step

1. ☐ Remove the bottom cover and the battery.
2. ☐ Remove every part listed below from the old upper case, in this order: battery, speakers, Wi-Fi card, display assembly, I/O board, system board (with world-facing camera).
3. ☐ Check what's pre-fitted on the new upper case, then transfer everything else.
4. ☐ Reconnect every cable, checking against your photos.
5. ☐ Reconnect the battery and test everything before closing up.

Common mistakes
✗ Losing track of which screw goes where.
✗ Forgetting a small part (tape, bracket, rubber feet).
✗ Closing up before testing everything.

If something goes wrong
Something doesn't work after → Check the photo of that area: a cable isn't plugged in.
Leftover screws → Find where they go before closing.

Verify before closing the ticket

- Every key passes the S2 keyboard test.
- Touchpad, display, camera, ports, speakers and Wi-Fi all work.
- No screws left over.

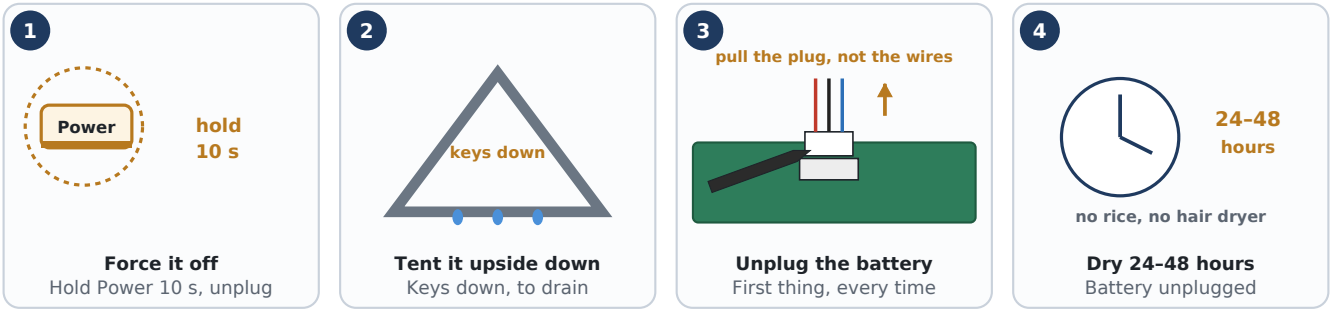
R17 Liquid spill response

LEVEL 3

Fixes	Any liquid on or in the device
Part	Often none; sometimes keyboard/top cover or system board
Time	10 min first response; 24–48 h drying
Tools	90%+ isopropyl alcohol, soft brush, lint-free cloths

In plain words. Water and electricity don't mix. The goal is to cut the power fast, drain the liquid, and let it dry completely before it's ever turned on again. Speed matters more than anything.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model

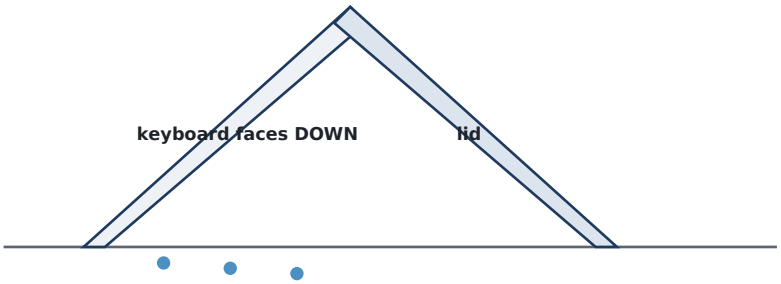


Illustration: drain position after a spill.

1. Power off (hold 10 s), unplug.
2. Tent upside down so liquid drains OUT.
3. Open and disconnect the battery ASAP.
4. Dry 24–48 h. No hair dryer, no rice.

Spill-resistant keyboards resist drips, not a full cup. Speed matters more than anything.

Do not power on to test. Current through liquid causes the corrosion that kills boards.

Step by step

1. ☐ Hold Power for 10 seconds to force off. Unplug.
2. ☐ Tent the device upside down (lid at 90°, keyboard down) to drain.
3. ☐ Open it and disconnect the battery as soon as possible.
4. ☐ Blot visible liquid; don't wipe it around.
5. ☐ Sugary or sticky spills: clean affected areas with a soft brush and isopropyl alcohol.
6. ☐ Dry open, battery disconnected, for 24 hours minimum (48 if soaked). No hair dryer, no rice.
7. ☐ Inspect for white or green corrosion; clean connectors.
8. ☐ Reconnect the battery, plug in, and run S2 fully.

Common mistakes
✗ Turning it on "just to check".
✗ Using rice or a hair dryer.
✗ Wiping liquid around instead of blotting.

If something goes wrong
White or green crust on the board → Corrosion: lead tech, Section 6.
Keys sticky afterwards → Keyboard repair.

Verify before closing the ticket

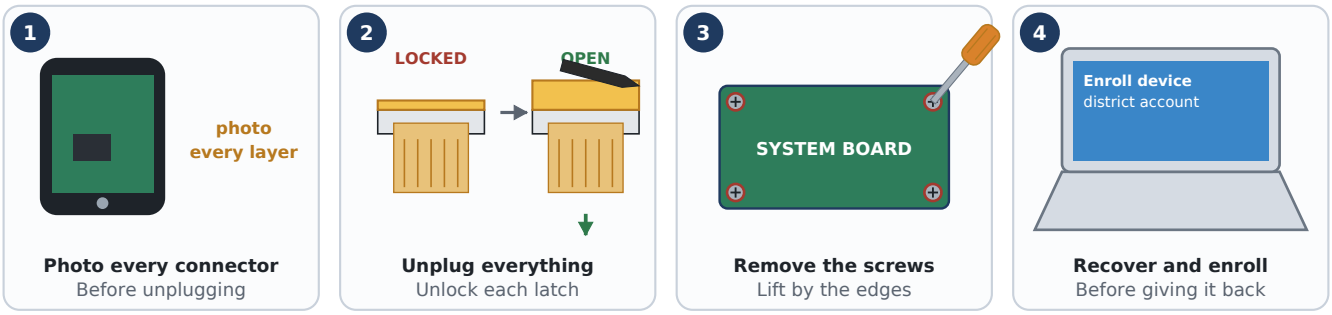
- All S2 tests pass.
- Sticky or dead keys: keyboard repair.
- Corrosion on the system board: Section 6.

R18 System board replacement (lead tech) LEVEL 3

Fixes	Dead USB-C ports, no power with a good battery, failed storage, RAM errors, failed recovery
Part	System board (by configuration; look up by serial)
Time	45–90 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, ESD strap

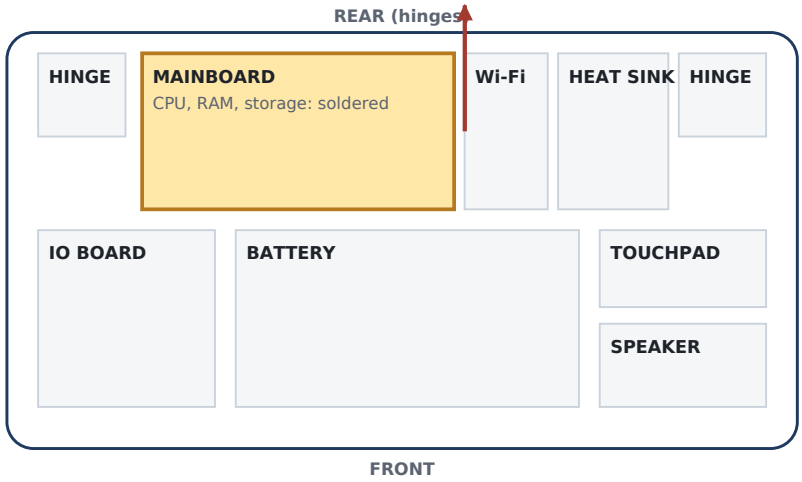
In plain words. The system board (main board) is the brain: processor, memory and storage are soldered onto it. Replacing it means unplugging everything, swapping the board, and setting the Chromebook up again. Only after every simpler fix has failed.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Battery out first (and the Wi-Fi card if it's socketed).
2. Unplug: display, I/O board, touchpad, keyboard, speaker, antennas
3. Lift by the edges once every screw and cable is free.
4. Lift by the edges onto the ESD mat.

Illustration: base interior, mainboard highlighted. Red arrow = lift side.

Identity and enrollment. Storage, serial number and enrollment live on the system board. The lead tech updates the asset record, deprovisions the old device in the Admin console if needed, and plans re-enrollment. Ask the manufacturer whether the new board ships with the serial set.

Confirm first with S1, S2, S3 and a connector reseat. Most "dead boards" are a battery, cable or connector.

Step by step

1. ☐ Remove the bottom cover, the battery, the Wi-Fi card, the heat sink.
2. ☐ Remove the display assembly and the Wi-Fi card.
3. ☐ Disconnect every cable from the board, including the world-facing camera (photograph first).
4. ☐ Remove the 8 M2×4 screws and lift the board out.
5. ☐ Fit the new board, reconnect every cable, refit the parts removed, then run recovery (S3) if needed and enroll.
6. ☐ Lenovo's HMM has the servicer write the serial and model data (VPD) to a new board with a repair shim. That's a depot step: ask Lenovo or district IT whether the replacement board ships programmed.

Common mistakes

✗ Replacing the board when a cable was the problem.
✗ Forgetting to move parts that stay (heat sink, SSD).
✗ Not updating the asset record.

If something goes wrong

Still dead with new board → Check the battery and charger first.
Won't enroll → Lead tech checks the Admin console.

Verify before closing the ticket

- Boots, enrolls and passes all of S2.
- Asset record and Admin console show the correct serial number.

6 Out of scope: escalate

These need soldering or board-level work, or parts that aren't sold. Tag the device with the ticket and send it to the lead tech.

Fault	Why	Route
USB-C port loose or dead	Usually soldered to the system board	System board or warranty
Storage or RAM failure	Soldered (system-board level)	System board or warranty
Stylus lost or dead	Not a repair	Order a replacement pen (look up by serial)
Burn marks, melted plastic, burnt smell	Fire hazard	Lead tech immediately
Any repair needing a soldering iron or hot air	Program policy	Lenovo warranty or lead tech

Lenovo support

- **Lenovo support:** pcsupport.lenovo.com (enter the serial number)
- **Parts lookup:** PartsLookup on pcsupport.lenovo.com (by serial)
- **Manual:** the Hardware Maintenance Manual (HMM) for your machine type, on pcsupport.lenovo.com

7 Sources

Checked October 2026. The procedures are the guide author's adaptation of these sources for a student bench; they are not manufacturer-issued instructions.

- Lenovo, *Lenovo 500e Chromebook Hardware Maintenance Manual*, January 2018 (copy at usermanual.wiki) — screw counts and removal order.
- Lenovo PC Support (pcsupport.lenovo.com) — parts lookup by serial.